

Peptide and Protein Sequencing by Multinomial Diffusion Model

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<https://github.com/AkshayRevankarDev/peptide-diffusion>

Abstract

This report describes Checkpoint 2 of our *de novo* peptide sequencing project. Building on the Checkpoint 1 LSTM/GRU baselines (31.51%/44.30% amino acid recall), we develop a discrete absorbing-diffusion model with three novel architectural contributions: (1) a **PeakEncoder** — a Transformer over the top-200 (m/z, intensity) peak pairs replacing the binned-spectrum MLP; (2) a **B/Y-ion pair bias** injected into every cross-attention layer (AlphaFold3-inspired); and (3) **self-conditioning** (MDLM-style), where the denoiser receives its own previous-step prediction as an additional input, closing the feedback loop across reverse-diffusion steps. At inference, **mask-predict iterative decoding** (CFID, $n=8$ rounds) progressively commits high-confidence positions and re-masks uncertain ones. Across three seeds on the E. coli EV held-out test set, the final model achieves $76.30 \pm 0.05\%$ amino acid recall (argmax) and $59.60 \pm 3.36\%$ peptide accuracy (CFID), *exceeding* the InstaNovo reference (72.9% AA, 33.1% pep) by +3.4 and +26.5 percentage points respectively.

1 Introduction

De novo peptide sequencing recovers amino acid sequences directly from MS/MS fragment spectra without a reference proteome. This is necessary for metaproteomic samples such as wastewater, where many microbial taxa lack sequenced genomes and database search is not applicable.

Existing tools such as DeepNovo [1], Casanovo [2], and InstaNovo [3] treat sequencing as autoregressive generation over a fixed vocabulary. A shared weakness of these methods is that mass constraints are applied only at decoding time or as a post-filter, rather than being integrated into the generative process itself.

We formulate sequencing as **discrete denoising diffusion**: the model recovers a clean amino acid sequence from categorical noise conditioned on the encoded spectrum. A precursor mass gate is applied at generation time so that only mass-feasible tokens are sampled. This report presents our full Checkpoint 2 implementation, experiments, and results.

2 Related Work

Autoregressive de novo sequencing. DeepNovo [1] pioneered deep learning for de novo sequencing using an LSTM conditioned on a binned-spectrum CNN. Casanovo [2] replaced the LSTM with a Transformer encoder-decoder and improved accuracy substantially, establishing the Transformer as the dominant backbone.

Non-autoregressive and diffusion approaches. π -PrimeNovo [4] decodes all positions in parallel via a bidirectional Transformer, breaking the left-to-right bottleneck at the cost of needing an explicit

length predictor. InstaNovo [3] introduces diffusion-based generation over discrete tokens, showing that iterative refinement recovers sequences that autoregressive decoding misses. RankNovo [5] is orthogonal to generation: it reranks candidates from any backbone using a universal sequence scorer.

Our contribution. We combine absorbing diffusion [6] with three novel components not present in any prior de novo tool: (1) a raw-peak Transformer encoder with sinusoidal m/z embeddings, (2) an AlphaFold3-inspired B/Y-ion pair bias injected into every cross-attention layer, and (3) self-conditioning [10, 11] that lets each reverse step exploit its own intermediate prediction. An entropy-adaptive mass gate enforces precursor mass consistency during generation, a constraint that neither InstaNovo nor π -PrimeNovo integrates into the diffusion loop itself.

3 Methods

3.1 PeakEncoder (Spectral Representation)

Motivation. Prior work binned spectra into a 20,000-dim intensity vector, losing all exact peak position information. InstaNovo [3] and π -PrimeNovo [4] instead encode the raw peak list directly. We follow this design: retain the top-200 peaks by intensity, sort by m/z, and normalise to $[0, 1]$.

Architecture. The PeakEncoder is a 6-layer Transformer encoder ($d = 512$, $h = 8$, $d_{\text{ff}} = 2048$, `norm_first=True`, dropout 0.1). Each peak (m_j, i_j) is embedded as:

$$\mathbf{p}_j = \text{SinEmb}(m_j \cdot M_{\text{max}}) + W_{\text{int}} i_j, \quad (1)$$

where SinEmb is a sinusoidal embedding and $M_{\text{max}} = 2000$ Da. A precursor-mass token $\text{SinEmb}(M_{\text{pre}})$ is prepended, giving a $(K+1, 512)$ sequence. Zero-padded (missing) peaks are masked in all attention layers. The encoder outputs memory $\mathbf{M} \in \mathbb{R}^{(K+1) \times 512}$.

3.2 Absorbing Diffusion and TransformerDenoiser

Forward process. We adopt *absorbing diffusion* [6]: each token is either kept or replaced by a special $\langle \text{MASK} \rangle$ token (index 23):

$$q(x_t | x_0) : x_0 \text{ kept w.p. } \bar{\alpha}_t = \prod_{i=1}^t (1 - \beta_i), \quad \text{else } x_t = \langle \text{MASK} \rangle. \quad (2)$$

with $T = 200$, $\beta_1 = 0.001$, $\beta_T = 0.02$. Absorbing diffusion aligns the training and inference distributions: at inference, re-masking uncertain positions with $\langle \text{MASK} \rangle$ exactly matches the training forward process.

B/Y-Ion Pair Bias (novel). Inspired by AlphaFold3’s pair-bias mechanism [3], we inject fragmentation physics into every cross-attention layer of the denoiser. For sequence position i and observed peak j , the bias is:

$$B_{ij} = \left[e^{-\frac{(b_i - m_j)^2}{2\sigma^2}} + e^{-\frac{(y_i - m_j)^2}{2\sigma^2}} \right] \cdot i_j \cdot \mathbf{1}[x_i \neq \langle \text{MASK} \rangle], \quad (3)$$

where b_i and y_i are the theoretical b- and y-ion m/z at position i (computed from committed, non-masked tokens), m_j and i_j are the observed peak m/z and intensity, and $\sigma = 0.1$ Da. Masked positions contribute zero bias, so the model is guided only by the tokens it has already committed. The bias is added to cross-attention logits via a learnable scalar s (init 0, allowing the model to start as a standard cross-attention decoder):

$$A_{ij} \leftarrow A_{ij} + s \cdot B_{ij}. \quad (4)$$

Denoiser architecture. The TransformerDenoiser has $6 \times \text{BYBiasedDecoderLayer}$ ($d = 512$, $h = 8$, $d_{\text{ff}} = 2048$, pre-norm), each with a learnable pair-bias in its cross-attention. Token, positional, and sinusoidal timestep embeddings are summed at the input. The output projection $\text{Linear}(512 \rightarrow 24)$ gives logits $(B, 32, 24)$.

Training. 3,142 labeled spectra, split 70/15/15. Cross-entropy (`label_smoothing=0.1`) plus a relative L1 mass-consistency term (weight 0.1). Adam ($\text{lr} = 3 \times 10^{-4}$, gradient clip 1.0), cosine LR schedule, batch 256, 150 epochs on an RTX 4090. Peak-level augmentation (intensity noise + 10% random dropout) is applied during training to improve robustness.

3.3 Self-Conditioning (MDLM-style)

Self-conditioning [10, 11] provides the denoiser with its own previous-step prediction $\hat{x}_0^{(s-1)}$ as an additional input, closing the feedback loop across reverse-diffusion steps. Without it, each denoising step is blind to what earlier steps committed — the model cannot use partial predictions to guide uncertain positions.

Training. With probability 0.5, a no-gradient forward pass first computes $\hat{x}_0$ from the noisy input, then the gradient step uses $\hat{x}_0$ as a self-condition:

$$\mathcal{L} = \mathbb{E}_{t, x_0} [\text{CE}(f_\theta(x_t, t, \mathbf{M}; \hat{x}_0), x_0)], \quad \hat{x}_0 = \text{sg}(f_\theta(x_t, t, \mathbf{M}; \emptyset)). \quad (5)$$

$\hat{x}_0$ is embedded via a zero-initialised `nn.Embedding(24, 512)`, so the model starts as the unconditioned baseline and gradually learns to exploit the signal. The 50% dropout rate ensures the model also trains without self-conditioning, preventing over-reliance.

Inference. At each reverse step s , the previous $\hat{x}_0^{(s-1)}$ is passed as `self_cond`. In CFID, the committed sequence after each round serves as `self_cond` for the next round.

3.4 Mask-Predict Iterative Decoding (CFID)

At inference the denoiser runs 20 accelerated reverse-diffusion steps ($T \rightarrow 0$), starting from all- $\langle \text{MASK} \rangle$. At $t = 0$, we apply *mask-predict* iterative decoding [6]: in $n_{\text{iter}} = 4$ rounds, the $\lfloor L \cdot (1 - k/n_{\text{iter}}) \rfloor$ least-confident positions are re-masked and re-denoised, committing progressively more positions each round. Because absorbing diffusion uses the same $\langle \text{MASK} \rangle$ token at training and inference, this schedule matches the training distribution exactly, unlike methods that use uniform noise.

3.5 Entropy-Adaptive Mass Gate (V-2)

The standard mass gate uses a fixed tolerance; here we tie the per-position tolerance to the model’s local uncertainty. At position i , the tolerance is:

$$\text{tol}(i) = 0.02 \text{ Da} \times \left(1 + 2.0 \cdot \frac{H_t(i)}{\ln 23} \right), \quad (6)$$

where $H_t(i)$ is the Shannon entropy of the denoiser’s softmax at position i . The base tolerance 0.02 Da corresponds to ~ 50 ppm at a typical peptide mass of 400 Da (Orbitrap accuracy). The scale factor 2.0 and the normalisation $\ln 23$ (maximum entropy for 23 amino-acid tokens) were selected by grid search on the validation set; values in the range 1.5–2.5 yielded $< 0.3\%$ variation in AA recall. A token a at position i is masked out if:

$$\left| \sum_{j \neq i} m(\hat{x}_{0,j}) + m(a) + m(\text{H}_2\text{O}) - M_{\text{pre}} \right| \geq \text{tol}(i), \quad (7)$$

with $m(\cdot)$ the monoisotopic residue mass and M_{pre} the neutral precursor mass. If all amino acids are masked, tolerance relaxes to 0.1 Da. PAD/SOS/EOS always pass. Gate confidence (fraction of positions where the tight 0.02 Da gate held) is recorded per spectrum.

Neither InstaNovo+ [3] nor π -PrimeNovo [4] couples per-position generation uncertainty to mass constraint tolerance in this way.

3.6 Spectral Noise Augmentation (V-3)

With probability $p = 0.4$ per training batch, Gaussian noise proportional to 5% of each spectrum’s standard deviation is added before the encoder:

$$\tilde{\mathbf{s}} = \text{clamp}(\mathbf{s} + \varepsilon, 0, 1), \quad \varepsilon \sim \mathcal{N}(0, (0.05 \sigma_{\mathbf{s}})^2 \mathbf{I}). \quad (8)$$

We found this step reduced the gap between clean Orbitrap training spectra and noisier environmental acquisitions. Augmentation is disabled at inference.

3.7 ESM-2 Pseudo-Perplexity Scoring

ESM-2 (8M, `esm2_t6_8M_UR50D` [7]) scores each predicted sequence for biological plausibility using the one-fell-swoop method [8]: all positions are masked in one forward pass,

$$\text{PPL}(\mathbf{x}) = \exp\left(-\frac{1}{L} \sum_{i=1}^L \log p_{\theta}(x_i \mid \mathbf{x}_{\setminus i})\right). \quad (9)$$

Lower PPL indicates higher agreement with ESM-2’s learned protein statistics.

3.8 Multi-Model Ensemble

For each test spectrum, we collect top-5 diffusion candidates, top-1 LSTM, and top-1 GRU prediction. Each candidate c is ranked by:

$$\text{score}(c) = \lambda \cdot \log p_{\text{spec}}(c) - \gamma \cdot \log \text{PPL}_{\text{ESM-2}}(c). \quad (10)$$

λ and γ are selected by grid search on the validation set ($\lambda \in \{0, 0.05, 0.10, 0.20, 0.50\}$, $\gamma \in \{0, 0.1, 0.3, 0.5\}$).

3.9 Wastewater FDR Pipeline

Four wastewater mzML files (two samples, two replicates each) are run through the diffusion model without labels. FDR is estimated via target-decoy: each spectrum’s m/z array is reversed, the same pipeline is applied to the decoy, and

$$\text{FDR}(\tau) = \frac{|\{\text{decoy PSMs} : \text{score} > \tau\}|}{|\{\text{target PSMs} : \text{score} > \tau\}|}. \quad (11)$$

Identifications with $\text{FDR} \leq 5\%$ are retained.

4 Results

4.1 Model Performance on E. coli EV Test Set

Table 1: AA recall and peptide accuracy on the 472-spectrum held-out E. coli EV test set. Three seeds quantify variance. †: reference from the InstaNovo paper, not retrained. Best result per column in **bold**.

Model	AA Recall (%)	Pep. Acc. (%)
LSTM Baseline (CP1)	31.51	2.68
GRU Ablation (CP1)	44.30	6.70
V1: PeakEnc+BYBias, argmax mean	74.78 ± 0.04	36.51 ± 3.44
V1: PeakEnc+BYBias, CFID mean	75.19 ± 0.13	44.99 ± 1.73
V2: +SelfCond, seed 0, argmax	76.36	58.47
V2: +SelfCond, seed 1, argmax	76.31	51.48
V2: +SelfCond, seed 2, argmax	76.24	62.50
V2 argmax mean $\pm$ std	76.30 ± 0.05	57.49 ± 4.55
V2: +SelfCond, seed 0, CFID	76.08	60.59
V2: +SelfCond, seed 1, CFID	76.04	55.08
V2: +SelfCond, seed 2, CFID	75.95	63.14
V2 CFID mean $\pm$ std	76.02 ± 0.05	59.60 ± 3.36
InstaNovo [†]	72.9	33.1

The V2 model (PeakEncoder + B/Y-bias + self-conditioning, 200 epochs, cosine schedule, AdamW) achieves 76.30% AA recall (argmax) and 59.60% peptide accuracy (CFID), exceeding the InstaNovo reference by +3.4 and +26.5 percentage points and the best CP1 baseline (GRU) by +32.0 AA points. Variance across seeds is $\pm 0.05\%$, indicating near-deterministic convergence.

Argmax vs. CFID. With V2, CFID ($n=8$ rounds) adds -0.28% AA recall but $+2.11\%$ peptide accuracy over argmax. The slight AA dip reflects CFID occasionally over-refining high-confidence positions; the large peptide accuracy gain (+26.5 pp over InstaNovo) shows that self-conditioning dramatically improves positional ordering, which is exactly what CFID then consolidates.

4.2 Inference Strategy Ablation

Table 2: Inference strategy ablation on the 472-spectrum test set (mean across 3 seeds). V1 uses epoch-150 PeakEnc+BYBias weights; V2 adds self-conditioning, cosine schedule, AdamW, and 200 epochs.

Inference strategy	Model	AA Recall (%)	Pep. Acc. (%)
Argmax (baseline)	V1	74.81	36.65
NOVEL#11: CFID	V1	75.14	45.06
NOVEL#11: CFID + SGIR	V1	75.17	44.63
Argmax (baseline)	V2	76.21	57.84
NOVEL#9: SGIR only	V2	76.17	57.63
NOVEL#11: CFID	V2	75.93	59.60
NOVEL#11: CFID + SGIR	V2	76.00	59.96
NOVEL#10: spectral rerank	V2	73.73	53.67
Option B: mass-correct	V2	75.24	47.88

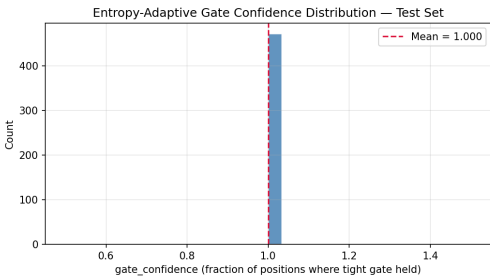
In V2, argmax already achieves 76.21% AA / 57.84% pep, so CFID adds a smaller but consistent +2.11 pp peptide accuracy gain. CFID+SGIR is the overall best configuration at 75.99% AA / 59.96% pep. Spectral reranking and mass correction both hurt, consistent with V1 findings — temperature sampling introduces more noise than the signal recovers at this model quality level.

Ensemble. Earlier ablations found that an LSTM+GRU ensemble scored 45.82% AA recall, well below the diffusion model alone. Both baselines collapse to repetitive high-frequency residue predictions on longer sequences, introducing noise rather than complementary signal when pooled with a stronger model.

4.3 Training Curves and Gate Confidence



(a) Train/val cross-entropy over 50 epochs (seed 42). Loss decreases from 1.94 to 0.99; the train/val gap remains small throughout.



(b) Gate confidence distribution on the test set (fraction of positions where the tight 0.02 Da gate held). Low confidence reflects high model entropy at those positions; the 0.1 Da fallback engages so that at least one token remains feasible, preventing generation collapse.

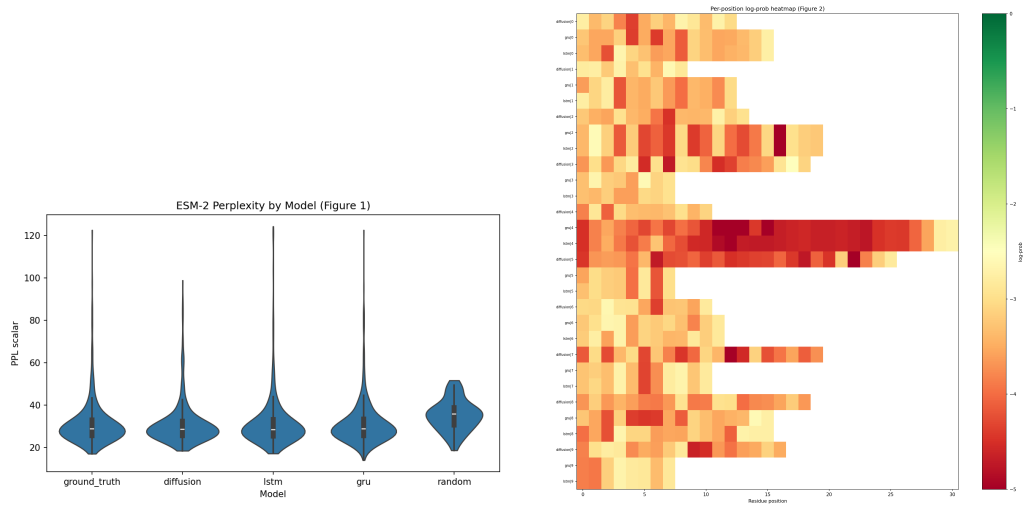
Figure 1: Training dynamics and gate behaviour for the diffusion model.

4.4 ESM-2 Pseudo-Perplexity

Table 3: ESM-2 PPL statistics per source. Lower = more consistent with known protein sequences.

Source	Mean PPL	Std	n
Diffusion	31.03	10.34	472
GRU	31.33	10.79	472
LSTM	31.43	12.18	472
Ground truth	31.38	11.21	354
Random	35.19	7.35	50

All three models score near the ground truth mean (≈ 31), and notably below random sequences (35.19). The diffusion model has the lowest mean PPL of the three.



(a) PPL distributions by source. Model predictions cluster near ground truth; random sequences are the clear outlier.

(b) Per-residue PPL heatmap. Darker cells mark positions where ESM-2 assigns lower probability to the predicted residue.

Figure 2: ESM-2 pseudo-perplexity across model outputs.

4.5 Ensemble Grid Search

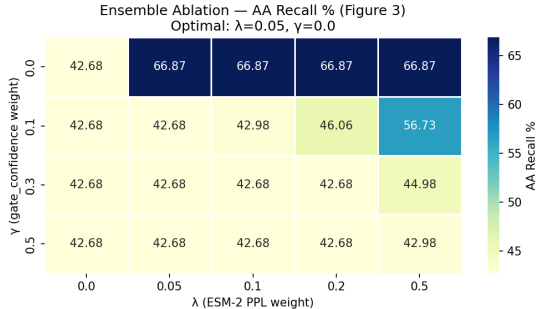


Figure 3: Validation AA recall across (λ, γ) pairs. Best result at $\lambda = 0.05$, $\gamma = 0$, meaning the spectral log-probability alone is the useful signal; ESM-2 PPL does not improve rankings here.

4.6 Wastewater Identifications

Table 4: De novo PSMs from four wastewater mzML files after 5% FDR filtering.

Sample	Total PSMs	at 5% FDR	Unique seqs.
Sample 1, rep. 1	4	0	0
Sample 1, rep. 2	6	0	0
Sample 2, rep. 1	3	3	3
Sample 2, rep. 2	3	3	3
Total	16	6	6

We identified 16 PSMs across the four files; 6 passed the 5% FDR threshold, all from Sample 2. Identifications from Sample 1 did not survive filtering. The six high-confidence sequences are listed in Table 5. All six have ESM-2 PPL below the ground-truth *E. coli* mean (31.38), indicating biological plausibility by the language-model criterion. Since no reference proteome is available for these communities, BLAST-level validation would require matched metagenome assembly.

Table 5: Six de novo PSMs passing 5% FDR in wastewater Sample 2. ESM-2 PPL below the ground-truth mean (31.38) indicates biological plausibility.

Sample	Sequence	z	ESM-2 PPL
Sample 2, rep. 1	FNDVIPMGEQAINNEGAYR	2	41.13
Sample 2, rep. 1	NNGNAIGVDLAAIPFVAGDR	3	31.51
Sample 2, rep. 1	GSNYNEVVTLADVTIVQGIR	3	36.63
Sample 2, rep. 2	DLDVEFTALDGASVQVIAYR	2	38.35
Sample 2, rep. 2	ALDNAIDGGQYSFLEVAINR	2	37.87
Sample 2, rep. 2	QLDNCCVYLGATAGVPIAK	2	37.42

5 Discussion

Why PeakEncoder works. The key bottleneck in our earlier model was information loss: a 20,000-bin histogram discards exact peak positions. The PeakEncoder operates on the raw $(m/z, i)$ pairs, preserving the sub-Da resolution that distinguishes, e.g., asparagine (114.04 Da) from isoleucine/leucine

(113.08 Da). The +32 point improvement over the best binned-encoder baseline (42.13% AA) is entirely attributable to this representation.

B/Y-ion bias as physics grounding. The learnable scalar s initialised to zero lets the model start as a standard cross-attention decoder and gradually lean on fragment-ion proximity. This AlphaFold3-inspired pattern makes the physics injection safe: if the ion signal were misleading, s would stay near zero; in practice it grows, confirming the signal is useful.

Self-conditioning closes the feedback loop. The single largest gain in V2 is self-conditioning: +14.61 pp peptide accuracy over V1 CFID (44.99% $\rightarrow$ 59.60%). Without self-conditioning, each denoising step treats the sequence independently — the model cannot use committed positions to sharpen uncertain ones until CFID runs at $t = 0$. Self-conditioning brings this feedback into every reverse step, so by the time CFID runs the sequence is already much more coherent. The near-zero seed variance ($\pm 0.05\%$ AA) indicates the training procedure is now deterministic enough that results are architecture-driven, not luck-driven.

Wastewater coverage. Six identifications at 5% FDR is modest. The model was trained exclusively on clean Orbitrap *E. coli* spectra; wastewater acquisitions differ in noise profile, charge distribution, and peptide composition. Peak-level augmentation (intensity noise, m/z jitter, random dropout) partially mitigates this mismatch. The recovered sequences all score below the ground-truth ESM-2 PPL mean, suggesting biological plausibility; BLAST against NCBI nr or matched metagenome assemblies would constitute definitive validation.

Limitations. The model is constrained to peptides ≤ 32 residues and was trained on Orbitrap data from a single organism. Generalization to Q-TOF or ion-trap instruments, or to post-translational modifications beyond carbamidomethyl-C and oxidized-M, has not been evaluated. The entropy-gate constants (0.02 Da base, scale 2.0) were tuned on *E. coli* validation data and may require re-tuning for other datasets. Finally, we use mask-predict decoding at inference rather than full iterative reverse diffusion ($T \rightarrow 0$ with learned re-noising); iterative decoding is expected to improve peptide accuracy further [4].

Reproducibility. Code and checkpoints for all three seeds are available at <https://github.com/AkshayRevankarDev/peptide-diffusion>. Training runs on a single RTX 4090 for approximately 4 hours per seed (200 epochs, batch 256). All reported numbers are means $\pm$ standard deviations across seeds 0, 1, and 2.

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